TR WAISWA
[illegible]
0748731657
M.A

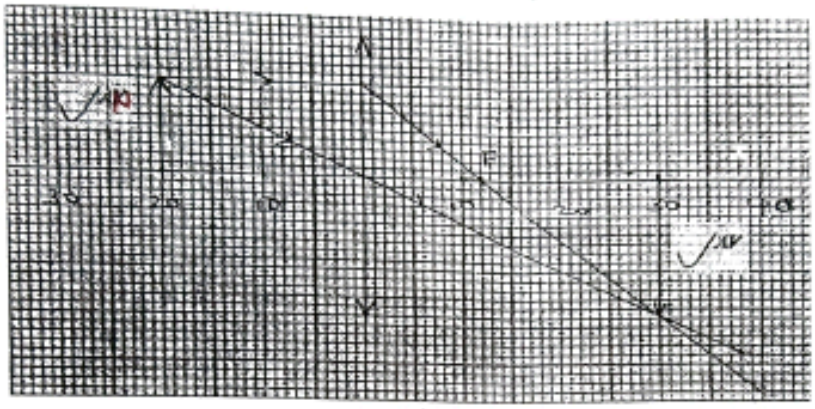
JINJA JOINT EXAMINATIONS BOARD

ITEM **JJEB UCE PHYSICS 535/1 AUGUST 2026 GUIDE** | **COMMENT**

Item One

(a) (i) **Suitability of the spectacles:**

Graphical Method Scale: 2 cm: 10 cm (*horizontal*)



Focal Length (*from graph*): $f = 12$ cm $= 0.120$ m AV

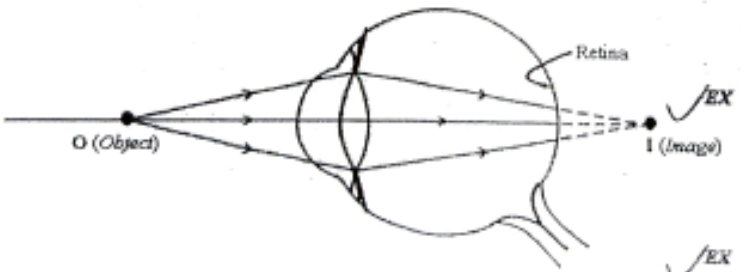
Lens Power, $P = {}^{1}/_{f} = {}^{1}/_{0.120} = +8.33$ **D** AV

Basing on scale drawing and calculations, since Power $P = +8.33$ **D** $\geq +8.29$ **D**, the spectacles can work for Jennie. AV

Type of lenses bought: Convex lenses (converging lenses) EX due to the positive value of the lens power. EX

Eye defect: Long Sightedness (Hypermetropia) EX

Light from a nearby object is brought to focus behind the retina. This occurs because either the eyeball is too short EX or the eye lens at its strongest setting is too weak.



EX EX

Correction: This defect is corrected by using a converging (convex) lens. The lens converges rays from a nearby object, making them appear to originate from further away, enabling the eye to focus the refracted rays directly onto the retina. EX

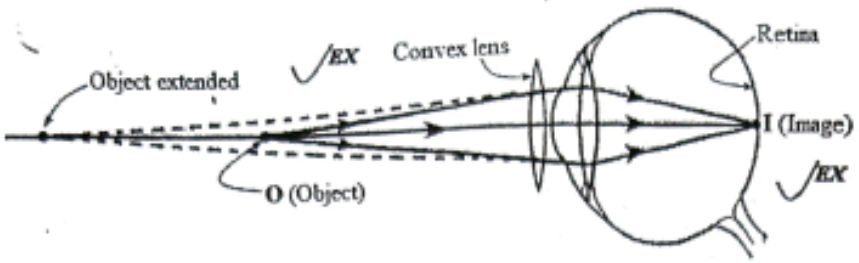
Qn 1

AN
≥7 ⇒ 06
4-6 ⇒ 05
2-3 ⇒ 04
1 ⇒ 03

Don't score for type of lens without reason. Qn 1

EX
≥7 ⇒ 06
4-6 ⇒ 05
2-3 ⇒ 04
1 ⇒ 03

APP
≥3 ⇒ 06
1-2 ⇒ 05



Why the disappearance of sound:

Sound is a mechanical wave; therefore, it requires a material medium for transmission. The pipe turned on by the doctor was connected to a vacuum pump connected to the sealed glass container, and when turned on, air was continuously sucked out of the container. As the air molecules gradually reduced, the intensity of the sound decreased. When the inside of the container became a complete vacuum, the sound disappeared entirely.

(ii) Testing the vaccines:

Wave frequency, f is given by $v = \lambda f$

(*where* $v = 340\ ms^{-1}$, $\lambda = 0.2$ m)

Thus, $340 = 0.2 \times f \Rightarrow f = 1700$ Hz

The frequency of sound produced by the monkeys is 1700 Hz < 1710 Hz. Therefore, the monkeys are healthy, and the doctor can proceed to test the vaccines.

(b) ***Side mirror in vehicles***: It provides a wide field of view and forms an upright image. This helps the driver see a broad range of traffic behind before attempting to overtake a vehicle.

Blind corner mirror on highways: A convex mirror is installed at sharp bends/corners on highways to help drivers see oncoming traffic from the blind side, allowing them to navigate the corner safely without causing accidents.

EX
≥7 ⇒ 6
4-6 ⇒ 5
2-3 ⇒ 4
1 ⇒ 3

WC	*EX*	15
	AN	12
WA	*AP*	06

Any two applications explained

Wave concept (WC)
wave application (

AP
≥3 ⇒ 6
1-2 ⇒ 5

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Item two

(a) **(i)** Light and darkness appearing on opposite sides of the earth occur due to the earth's rotation[EX] about its axis as it orbits the Sun.
As the earth rotates, one faces[EX] the sun making it to receive direct sunlight[EX], hence light on that side, while the other side faces away[EX] from the sun does[EX] not receive sunlight, hence being in darkness.

(ii) **Seasonal variation**

The difference in seasons is caused by the earth's[EX] tilt at an angle of 23° from its axis of rotation, combined with its revolution[EX] around the Sun.
As the earth revolves, one hemisphere tilts toward the sun and the other tilts away from the sun.[EX]
The island was in the hemisphere tilted towards the sun, receiving higher[EX] solar intensity and so making it to be in the summer.[EX]
The tourist's hemisphere was tilted away from the sun, receiving lower[EX] solar intensity and so making it to be in winter.[EX]

Fluctuation of water levels

The rise and fall of water level is termed as tides which are caused by the gravitational[EX] forces of the moon and sun on the earth's water bodies.
The moon's gravity pulls[EX] the water towards it, creating a bulge (high tide)[EX] on the side of the earth facing it.
On the side of the earth opposite to the moon, the earth-moon rotation around a common[EX] center creates centrifugal[EX] force which causes another bulge (high tide).[EX]
As the earth spins, its different locations pass through these bulges, experiencing high and low tides.

(b) The Citymapper App uses Global Positioning System[AP] (GPS) to help in navigation
A network[AP] of satellites orbiting the earth broadcast signals containing their location, and the exact time the signal was sent. The GPS in the phone detects[AP] the signals from the four satellites and measures the time and distance each signal takes to reach it from each satellite via trilateration.[AP]

Qn 2

EX
≥ 11 ⇒ 06
6-10 ⇒ 05
3-5 ⇒ 04
1-2 ⇒ 03

AP
≥ 3 ⇒ 06
2 ⇒ 05
1 ⇒ 04

The point where the distance spheres from the satellites intersect gives the phone's precise location on earth.
The Citymapper App places the calculated location on its digital map and compares it with the stored map data to identify the destination.
As the person moves, the phone repeatedly receives new GPS signals, re-calculates the location every few seconds in real time and provides the direction to the destination.

Benefits of Uganda's investment in space exploration

- improved weather forecasting and climate monitoring
- enhanced disaster preparedness and environmental management
- upgraded national telecommunications and connectivity
- advanced transportation navigation.
- promotion of scientific research, innovation, and STEM education
- job creation, technological industrialization, and economic expansion
- strengthened national security and defense surveillance capabilities

SC	EX	18
	AN	00
SA	AP	12

Any three benefits stated

Qn 3
EX
≥3 ⇒ 06
2 ⇒ 5
1 ⇒ 4
AN
≥9 ⇒ 06
5-8 ⇒ 05
1-4 ⇒ 4
APP
≥3 ⇒ 06
2 ⇒ 05
1 ⇒ 03

Item three

(a) (i)

Production timeline

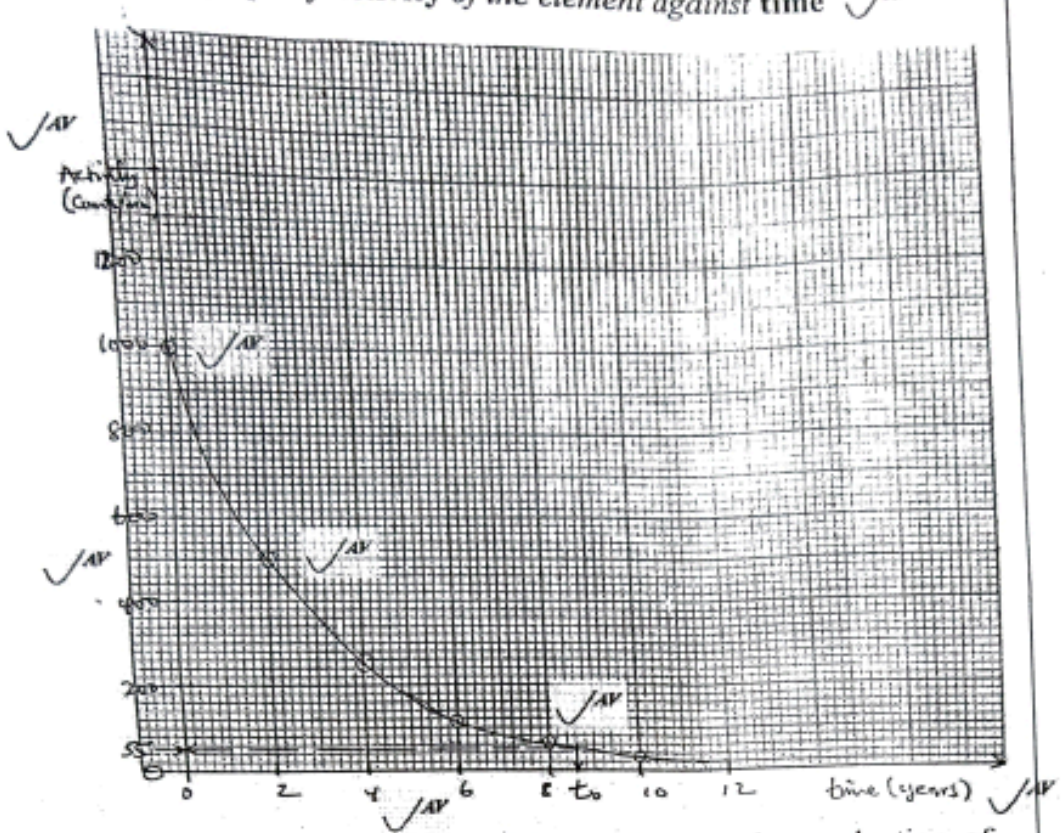
The table below shows the decay process of the installed element.

New activity	1000	500	250	125	62.5	31.25
Time taken to decay	0	2	4	6	8	10

Activity for production = Background activity + 10
= 45 + 10
= 55 counts min per minute

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A graph of **activity** *of the element against* **time**



From the graph, the ministry will start the production of electricity after 8.6 years.

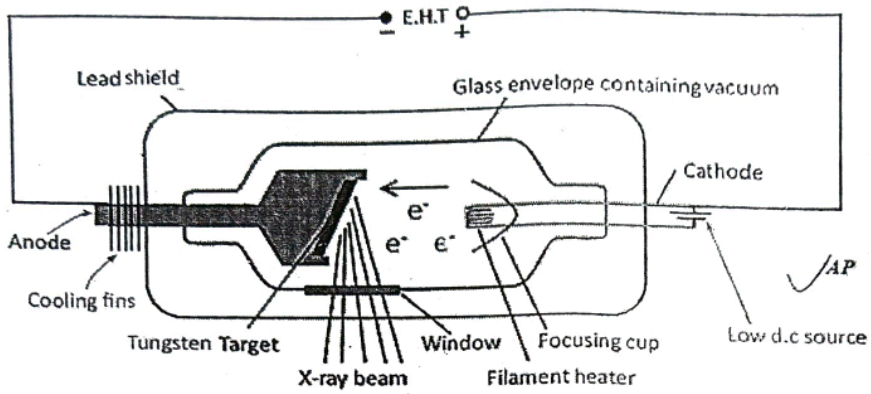
(ii) **Generation of electricity in a nuclear reactor.**

Uranium is a highly unstable radioactive element. It is formed into small fuel pellets and stacked inside long tubular fuel rods, which are assembled into the reactor core.
When a neutron strikes a uranium nucleus, the nucleus becomes more unstable and splits by the process called nuclear fission, releasing heat energy and three additional neutrons. The released neutrons strike nearby uranium nuclei, triggering further fission events and creating a chain reaction that generates immense heat.
Control rods (made of boron) are inserted or withdrawn from the core to absorb excess neutrons so as to regulate the fission rate. The rods are surrounded by water to act as a coolant, pumped into the core. The extreme heat transforms the water into high-temperature and high-pressure steam.
This steam spins the turbines which in turn drives the electric generator, hence producing electricity.

Economic effect: Provides reliable electrical energy to power local industries, generate employment, and accelerate infrastructural growth in the community. ✓EX

Health Risks: Presents potential hazards from exposure to ionizing radiation, which can lead to cellular damage, genetic mutations, and cancer development in humans. ✓EX

(b) The device used is an **X-ray** tube.



✓AP

Operation:

A low voltage D.C. source heats ✓AP the filament which releases the electrons by thermionic emission.

An E.H.T supply creates a strong ✓AP electric field that accelerates ✓AP electrons toward the tungsten target on the copper anode.

Upon striking the target, 99% of the kinetic energy is converted ✓AP into heat, which is safely conducted away by copper anode, through the cooling ✓AP fins. Only about 1% is converted ✓AP into X-rays.

Application:

The X-rays emerge through a restricted aperture ✓AP in the protective lead shield toward the patient's arm.

Soft tissues and flesh allow X-rays to pass ✓AP easily, creating a dark ✓AP exposure area on the underlying photographic plate.

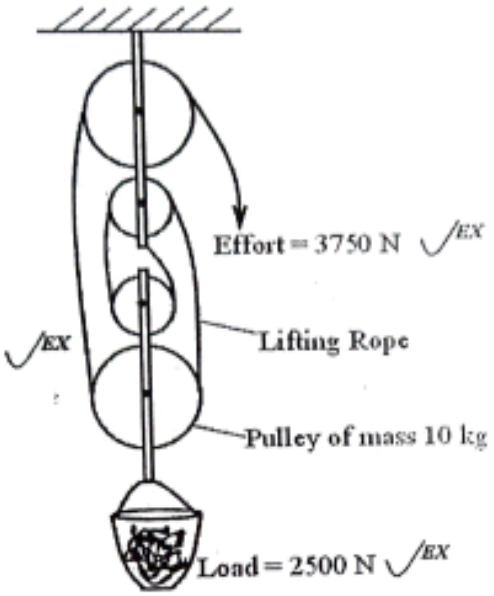
Dense bones absorb ✓AP the X-rays, leaving a light/white ✓AP image on the plate. Fractures or breaks appear as dark ✓AP lines within the bone structure, showing the exact location of the fracture.

PC	*EX*	11
	AP	13
PA	*AP*	13

Item four

(a) **Chief Engineer's request**:

Pulley System Arrangement:



$V.R. = 4$, $M.A. = \frac{L}{E}$.

Load (L): $L = (250 \times 10) + (20 \times 10)$
$= 2500 + 200 = 2700$ N AV

Effort (E): $E = 3750$ N, $M.A. = \frac{2700}{3750} = 0.72$ AV

Efficiency (η): $\eta = \frac{M.A.}{V.R.} \times 100 \Rightarrow \eta = \frac{0.72}{4} \times 100 = 18\%$ AV

Inclined ramp system:

Ramp length (l): $l = \frac{14}{\sin 38^\circ} = 22.74$ m AV

Work Output (W_O): $W_O = mgh = 250 \times 10 \times 14$
$= 35000$ J AV

Work Input (W_I): $W_I = F \times l = 3750 \times 22.74 = 85275$ J AV

Efficiency (η): $\eta = \frac{W_O}{W_I} \times 100\% = \frac{35000}{85275} \times 100\% = 41\%$ AV

Since the efficiency of the pulley system is $18\% < 71\%$ and that of the inclined plane is $41\% < 71\%$, AV none of the systems meets AV the chief engineer's requests.

Joby's ability

Using $s = ut + \frac{1}{2}at^2$, where $u = 0\ \text{ms}^{-1}$, $t = 10\ \text{s}$, $s = 20\ \text{m}$
$\Rightarrow 20 = (0 \times 10) + \frac{1}{2} \times a \times 10^2$ and therefore, $a = 0.4\ \text{ms}^{-2}$ ✓AV
From $v = u + at$: $v = 0 + (0.4 \times 10) = 4\ \text{ms}^{-1}$ ✓AV
Since Joby's velocity in 10 s is $4\ \text{ms}^{-1} < 4.3\ \text{ms}^{-1}$, he will not manage the task assigned to him by the chief engineer. Therefore, an automated machine system should be used. ✓AV

Final temperature of the mixture

Assuming no heat loss to the surroundings:
Heat lost by the pulley = Heat gained by ice + Heat gained by water + Heat gained by container ✓AV

$$\Rightarrow (10 \times 490 \times (110 - \theta)) = \left(\frac{231}{1000} \times 3.36 \times 10^5\right) + \left(\frac{231}{1000} \times 4200 \times \theta\right) + (3 \times 2100 \times \theta)$$ ✓AV

$$\Rightarrow 4900(110 - \theta) = 77616 + 970.2\theta + 6300\theta$$
$$\text{Thus, } 539000 - 4900\theta = 77616 + 7270.2\theta$$
$$\Rightarrow 12170.2\theta = 461384$$
$$\therefore \theta = 37.9°\text{C}$$ ✓AV ✓AV

Since the final temperature of the pulley is $37.9°\text{C} > 37.8°\text{C}$, Joby should add a few ice blocks. ✓AV

Joby's different body feelings

When Climbing: The gravity of the Earth acts on Joby in the opposite direction. Joby therefore does work to overcome the force of gravity, which results in energy loss, hence feeling tired.

Lowering the Pulley: Work is done by gravity on the pulley, causing it to move under free-fall principles, so Joby does not feel tired.

Cool air blowing towards Joby: Land has a smaller specific heat capacity than water. During the daytime, land becomes hot faster than water. This causes the temperature of air close to the land to increase, making it less dense so that it rises. The cooler, denser air over the lake blows towards the land to replace the rising hot air, causing cool air to blow towards Joby.

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Qn 4

EX
≥6 ⇒ 6
4-5 ⇒ 5
2-3 ⇒ 4
1 ⇒ 2

MC_1	EX	14
	AN	20
MA_1	AP	04

AN
≥10 ⇒ 6
7-9 ⇒ 5
3-6 ⇒ 4
1-2 ⇒ 2

AP
≥3 ⇒ 6
2 ⇒ 5
1 ⇒ 4

(b) Water has a higher specific heat capacity than oil. This allows it to absorb larger quantities of heat from car radiators without undergoing high temperature changes or evaporating easily. Oil heats up and evaporates over much smaller temperature changes.

Item five

(a) (i) Nelkon's observation:

Before the bullet is fired initially, the gun and the bullet are at rest. When fired, the expanding gas from the gunpowder pushes the bullet forward out of the barrel with forward momentum. By *Newton's third law of motion*, for momentum to be conserved, the gun moves backward with an equal and opposite momentum, causing the recoil/backward motion observed by Nelkon.

Jalia's observation:

When the nozzle of an inflated balloon is opened, the stretched rubber pushes the air backward. *By Newton's third law of motion*, the backward push of the escaping air creates an equal forward push that drives the balloon in the opposite direction.

(ii) Maximum height reached by the ball

Given: $t = 1.2$ s, $g = 10\ \text{ms}^{-2}$, $v = 0\ \text{ms}^{-1}$

Using $u = v + gt$: $u = 0 + (10 \times 1.2) = 12\ \text{ms}^{-1}$

Height reached from throw point (h):

$$h = \frac{u^2 - v^2}{2g} = \frac{12^2 - 0}{2 \times 10} = 7.2\ \text{m}$$

Total height above ground (H):

$$Jalia's\ height = \frac{140}{100} = 1.4\ \text{m}$$

$$H = 7.2 + 1.4 = 8.6\ \text{m}$$

Since 8.6 m < 10 m, Jalia did **not** win the prize.

What to do next time:

To reach 10 m, the required height from throw point is $h = 10 - 1.4 = 8.6$ m. From $h = \frac{u^2 - v^2}{2g}$

$$8.6 = \frac{u^2 - 0}{2 \times 10} \Rightarrow u^2 = 172 \quad \therefore \quad u \approx 13.1\ \text{ms}^{-1}$$

Qn 5

EX

≥5 ⇒ 06
3-4 ⇒ 05
2 ⇒ 04
1 ⇒ 2

AN

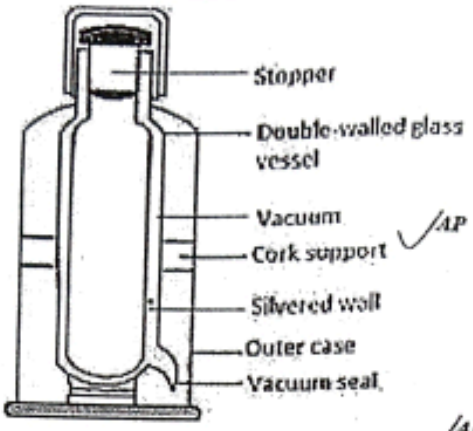
≥6 ⇒ 06
4-5 ⇒ 05
2-3 ⇒ 4
1 ⇒ 2

APP

≥ 6 ⇒ 06
4-7 ⇒ 5
1-3 ⇒ 4

She must throw the ball with an initial velocity greater than 13.1 ms^{-1} in order to win. AP

(b) Vacuum flask and heat loss control

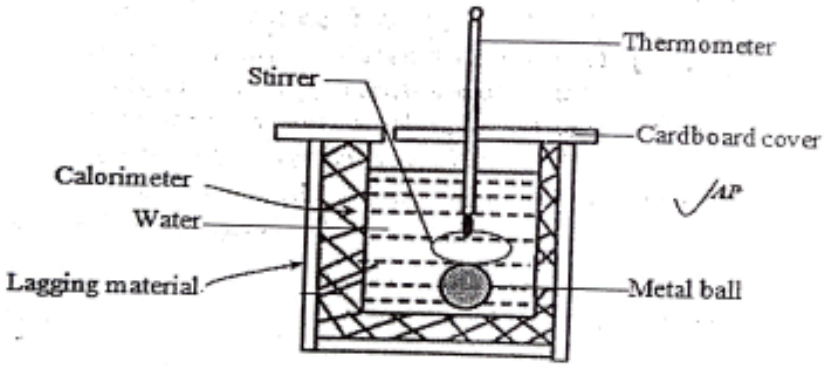


AP

Vacuum space between the double silvered walls minimizes heat loss by conduction and convection because there are no particles to transfer heat.
Double-silvered surfaces reflect heat radiation back into the flask, minimizing radiation loss. AP
Insulating stopper / Cork seals the top, preventing heat loss by convection and evaporation. AP
Vacuum seal prevents air from entering the space between the walls. AP

MC_2	EX	10
	AN	12
MA_2	AP	15

Experimental determination of specific heat capacity of a metallic ball using the method of mixtures



AP

Water of mass, m_w, and specific heat capacity c_w is poured gently into a calorimeter of mass, m_c and specific heat capacity, c_c. AP

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The initial temperature, θ_1 of water and the calorimeter is read from the thermometer and recorded.
The hot metal ball at a temperature of 100°C and of mass, m_b whose specific heat capacity, c_b is to be determined, is transferred quickly from the flask to the water in the calorimeter.
The calorimeter is then covered and the mixture in it is stirred gently.
The final temperature, θ_2 of the calorimeter and its contents is read from the thermometer and recorded.
The specific heat capacity, c_b of the metal ball is then obtained from the expression:

$$c_b = \frac{m_w c_w(\theta_2-\theta_1)+m_c c_c(\theta_2-\theta_1)}{m_b(110-\theta_2)}$$

Item six

(a) (i) **Cost of electricity**

Cost = power (kW) × ttime (h) × unit cost

Power for **A:** $P_A = \frac{360\text{ W}}{1000} = 0.36\text{ kW}$

Power for **B:** $P_B = \frac{V^2}{R} = \frac{72^2}{30} = 172.8\text{ W} = 0.1728\text{ kW}$

Power for **C:** $V_C = 240 - 120 = 120\text{ V};$

$$P_C = \frac{120^2}{2000} = 72\text{ W} = 0.072\text{ kW}$$

Power for **D:** $P_D = 115.2\text{ W} = 0.1152\text{ kW}$

Total Power = 0.36 + 0.1728 + 0.072 + 0.1152 = 0.72 kW

Cost = 0.72 kW × (8 × 30) × 9.6 = 1658.88 **US dollars**

Since the total cost is 1658.88 **US dollars** > 1655 **US dollars**, then Johnson should not be refunded the rent.

Burning of appliances and corrective measures

In the circuit diagram, some appliances are connected in series with the mains and each other. When there is a sudden increase in circuit voltage, current increases rapidly. Resistance causes a severe heating effect, leading to high temperatures and burning out appliances.

The appliances are connected in parallel and a circuit breaker or fuse installed to break the circuit in case there is excessive current.

Qn 06

EX

	W. Scores
7/5	06
3-6	05
1-2	04

AN

≥ 6 ⇒ 06

4-5 ⇒ 05

1-3 ⇒ 04

APP

≥ 5 ⇒ 06

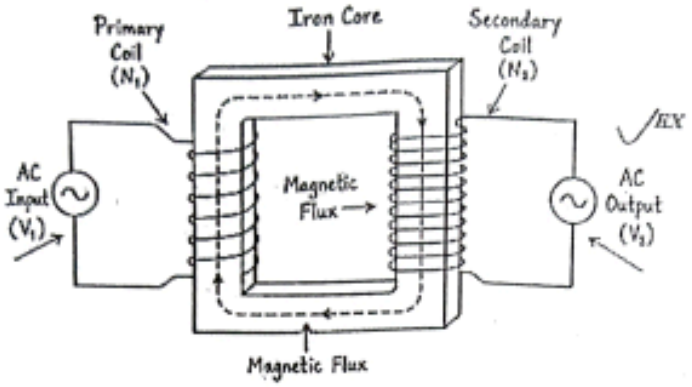
3-4 ⇒ 05

1-2 ⇒ 04

(ii) **Current *vs* step-up transformer requirement**

Using $P = IV$; $360 = I \times 120 \Rightarrow I = 3$ A ✓AV
Since source current is 3 A < 3.5 A, ✓AV

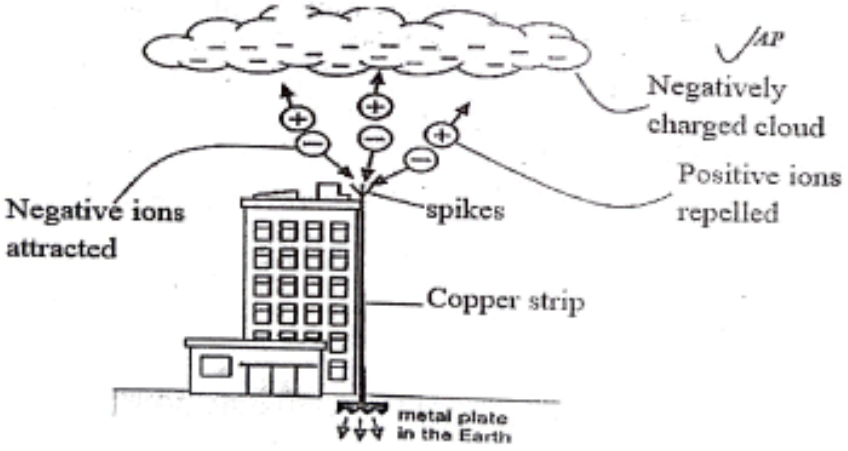
✓AV
Johnson needs to install a **step-up transformer** to boost voltage where required.



✓EX
The a.c voltage on primary coil creates a changing magnetic field in the soft iron core. This changing field induces a changing voltage in the secondary coil by electromagnetic induction (Faraday's law). Voltage in the secondary coil is directly proportional to the number of turns: $\frac{V_2}{V_1} = \frac{N_2}{N_1}$. Since the secondary coil has more turns, the voltage increases (stepped up).

(b) **Protection against lightning** ✓AP

A **lightning conductor** is installed on the rooftop.



When a negatively charged cloud passes over a building with a lightning conductor, it induces positive charges on the sharp spikes of the conductor.

FC_1	EX	09
	AP	11
FA_1	AP	09

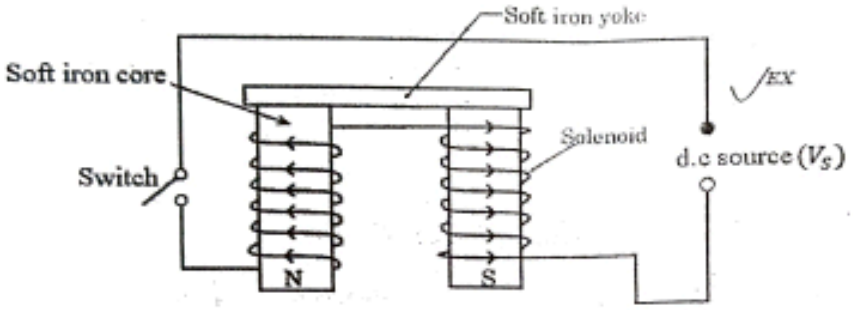
If diagram not drawn, distribute the score to the explanation

Due to high charge concentration a strong electric field is generated around the spikes, ionizing nearby air molecules, hence creating into ion pairs.
Positive ions are repelled toward the cloud to neutralize some of the negative charges on the clouds
Negative ions are attracted to the spikes, and neutralize the induced positive charges of the spikes, but the excess charge safely conducts to the ground through a thick copper strip.
This lowers the potential difference between cloud and building, and so preventing dangerous strikes, hence safeguarding the building.

Item seven

(a) (i) **Electromagnetic scrap-lifter design**

To lift a large amount of metal scrap requires a strong electromagnetic metal scrap lifter



An insulated copper wire is wound around both arms of a U-shaped soft iron core in opposite sense to form a solenoid. When the switch is closed, D.C. current flows through the solenoid. A magnetic flux is generated, which links up with the arms of the soft iron core.
The arms of soft iron core become magnetized strongly with opposite poles at the ends and attract magnetic scrap metal by induction. The metal scrap is thus transferred to the crushing machine.

Qn 07

EX
≥ 7 ⇒ 06
3-6 ⇒ 05
1-2 ⇒ 04

AN
≥ 3 ⇒ 06
2 ⇒ 05
1 ⇒ 4

APP
≥ 5 ⇒ 06
3-4 ⇒ 05
1-2 ⇒ 04

System reliability.

Assuming an ideal transformer:

$N_p = 100, N_s = 332, V_p = 125\ \text{V}$ ✓AV

$$\frac{V_p}{V_s} = \frac{N_p}{N_s} \Rightarrow \frac{125}{V_s} = \frac{100}{332} \Rightarrow V_s = 415\ \text{V}$$ ✓AV

Equivalent Load Resistance (R_L):

$$\frac{1}{R_L} = \frac{1}{72} + \frac{1}{30} + \frac{1}{\cancel{108}\ 10} \Rightarrow R_L \approx 10.286\ \Omega$$ ✓AV

Secondary Current (I_s):

$$I_s = \frac{V_s}{R_L} = \frac{415}{10.286} = 40.35\ \text{A}$$ ✓AV

✓AV ✓AV

Since $40.35\ \text{A} > 14.27\ \text{A}$, the system is reliably efficient.

Why A.C to D.C

✓EX

Soft iron is a temporary magnetic material. When alternating current is passed through the solenoid, the induced magnetic flux starts fluctuating ✓EX continuously. This makes the polarity of the core to reverse rapidly and become weak.

The rapidly reversing ✓EX polarity causes the electromagnet to lose its holding ✓EX grip on the metal scrap.

System modification for high performance

- Increase the number of turns in the secondary coil of the transformer to boost current. ✓EX
- Increase the number of turns in the solenoid coils. ✓EX
- Reduce circuit resistance.
- Use high-permeability soft iron cores to maximize induced magnetic strength.

Score any two modifications

(b) **Heat generation in the wires**

When the voltage is applied across the wire, free electrons in the wire are accelerated through the metal lattice. The moving electrons then collide with fixed lattice atoms, losing some of their kinetic energy to the atoms. This increases the internal energy of the lattice atoms, making them to vibrate with higher frequency and amplitude. As the atoms vibrate, they rub against each other, hence production of heat in the wire

FC_2	*EX*	13
	AP	06
FA_2	*AP*	10

Other applications of electromagnets

Relay Switches: Safe operation of high-power industrial circuits.
Electric Bells: Used for signaling session changes in schools.
Transformers: Stepping up or stepping down A.C. voltage for efficient long-distance power distribution.

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END